Janus RAG — A Human's Guide

Status: LOCKED v1.3 — 2026-06-07
Purpose: Understand every component of the Riva + Janus RAG architecture
Style: Plain language, analogies, examples, zero jargon without explanation

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Prologue: What Problem Are We Solving?

Riva is an AI assistant for Naga City. Citizens ask it questions like:

> "Magkano ang building permit fee?" (How much is the building permit fee?)
>
> "Ano ang requirements ng business permit?" (What are the requirements for a business permit?)
>
> "Bakit kailangan ng barangay clearance?" (Why do I need a barangay clearance?)

These seem like simple questions. But behind each one is a web of interconnected information spread across hundreds of PDFs — ordinances, citizen charters, budget reports, bid results, downloadable forms. Some are scanned photocopies from 1998. Some are clean 2025 PDFs. Some contain legal amendments that change fees over time.

The challenge: How do you build one system that answers ALL of these correctly?

The answer is **Janus RAG** — a system that uses TWO different ways of finding information, each specialized for a different type of question. Think of it like having both a librarian AND a search engine. The librarian knows exactly where every book is and how they relate to each other. The search engine can find any mention of a topic across thousands of pages. Together, they cover each other's blind spots.

Chapter 1: The Two Brains

Janus RAG has two retrieval systems. Let me introduce them.

Brain 1: The Graph (The Librarian)

The graph is a **knowledge graph** — a database that stores information as nodes (things) connected by edges (relationships). It's like a giant bulletin board with pushpins and strings connecting them.

The graph stores **14 types of things** (we call them nodes):
- JURISDICTION (the territory — Naga City)
- LGU (the government — City Government of Naga)
- OFFICE (departments — City Treasurer, City Engineer)
- SANGGUNIAN (the city council)
- LEGISLATION (ordinances, resolutions, executive orders)
- SERVICE (things citizens can do — "Apply for Building Permit")
- REGULATION (permits, licenses, fees — "Building Permit", "Business Tax")
- FUND (money — General Fund, Development Fund)
- EXTERNAL_AGENCY (national agencies — BFP, DOLE, DENR)
- EXTERNAL_ENTITY (businesses, contractors, banks, citizens)
- SESSION (council meetings)
- COMMITTEE (council committees)
- PROCUREMENT (bidding projects — "Buy 2 dump trucks")
- ZONING_CLASSIFICATION (land zones — residential, commercial, industrial)

And **37 types of relationships** (we call them edges) connecting these things. When someone asks a structural question — "how much?", "what are the requirements?", "who issued this?" — the graph can follow the strings and give a precise answer.

The graph's strength: When it knows something, it knows it with absolute precision. It can tell you not just the fee amount, but which ordinance authorized it, when it took effect, and whether it's still current.

The graph's weakness: It only knows what we've explicitly mapped into it. If a citizen asks about something we haven't mapped — say, the history of the Peñafrancia Festival — the graph has nothing to say.

Brain 2: The Vector Index (The Search Engine)

The vector index is a **semantic search engine**. It takes every document we have — ordinances, citizen charters, budget reports, web pages, tourism posts — splits them into small chunks, and converts each chunk into a list of numbers (called an "embedding") that represents what the text is about, not just what words it contains.

When someone asks a question, the system converts the question into the same kind of number list and finds the chunks whose numbers are closest — meaning they're about the same topic, even if they use different words.

The vector index's strength: It can find information across everything we've ever fed it. If you ask about the history of Naga City, it searches all documents and finds the tourism page that mentions the city's founding. It doesn't need us to have mapped "city history" into the graph.

The vector index's weakness: It can't do precision. If you ask "how much is the building permit fee?", the vector index might return a chunk from an old 2018 PDF that says ₱3,500 — even though the current fee is ₱5,000. It has no concept of "current vs. outdated." It just finds similar text. It also can't follow chains of requirements — it might find the building permit document but miss that you also need a fire safety certificate from BFP.

Chapter 2: The Router — Who Handles What?

When a citizen types a question, the first thing Janus RAG does is figure out WHICH brain should handle it. This is the **Intent Classifier**.

How the Router Works

Imagine a receptionist at an office building. When you walk in, they ask what you need. If you say "I'm here to pay my taxes," they send you to the Treasury window. If you say "I want to understand why we have zoning laws," they send you to the information desk.

The Intent Classifier is our receptionist. It scans your question for trigger words and assigns TWO scores:

Score 1: How STRUCTURAL is this question? (Does the answer live in the graph?)
- High structural: "magkano" (how much), "sino" (who), "kailangan" (requirements), "ordinance number"
- These questions need precise, authoritative answers with legal citations

Score 2: How NARRATIVE is this question? (Does the answer live in documents?)
- High narrative: "bakit" (why), "history," "explain," "ano ang ibig sabihin" (what does it mean)
- These questions need context, background, and explanation

The router also determines WHO you are:

Citizen queries: "Magkano ang fee?", "Ano requirements?", "Paano mag-apply?"
Employee queries: "Magkano nakolekta?" (How much was collected?), "Status ng procurement?"

This matters because employees get access to more of the graph (financial data, procurement trails, draft legislation) while citizens see only what's relevant to them.

The Routing Decision

Based on these scores, the question goes to one of three destinations:

Structural Score	Route	Meaning
≥ 0.7	**Graph Primary**	The graph handles this. Vector is on standby.
0.4–0.6	**Both**	Run both brains. The merger decides what to use.
≤ 0.3	**Vector Primary**	The vector index handles this. Graph provides legal citations if available.

Before Routing: The Synonym Translator

Filipinos don't always use official terminology. A citizen might say:
- "bgy clearance" instead of "Barangay Clearance"
- "pahintulot magtayo" (Bikol-influenced) instead of "Building Permit"
- "mayor's permit" instead of "Business Permit"
- "hm po" (text-speak for "how much")

Before the router even looks at the question, a **Synonym Alignment Interface** translates these colloquial terms into the canonical names that the graph knows. Think of it like a bilingual receptionist who understands both formal government language and how regular people actually talk.

Chapter 3: The Graph Path — How the Librarian Thinks

When a question is routed to the graph, it goes through **five sequential checks** — like security checkpoints at an airport. Each checkpoint ensures the information is valid, current, and complete.

Checkpoint 0: Is the Library Even Open?

```
TRY: Connect to KuzuDB (the graph database)
CATCH: If it's down → Switch to vector-only with a disclaimer:
       "The graph database is temporarily unavailable.
        Answers are from document search only.
        Verify all fees with the issuing office."
```

This is like the librarian calling in sick — we still have the search engine, but we tell people the answers might not be as precise.

Checkpoint 1: Temporal Filter — Is This Information Still Valid?

Laws change. Fees increase. Ordinances get repealed. The graph needs to know what's current.

Every piece of information in the graph has three dates:
- **effective_date:** When it became active

- **repeal_date:** When it expired (or null if it's still active)
- **publication_date:** When it was officially published (laws take effect 15 days after publication)

Checkpoint 1 asks: "Is this information valid at the date the citizen is asking about?"

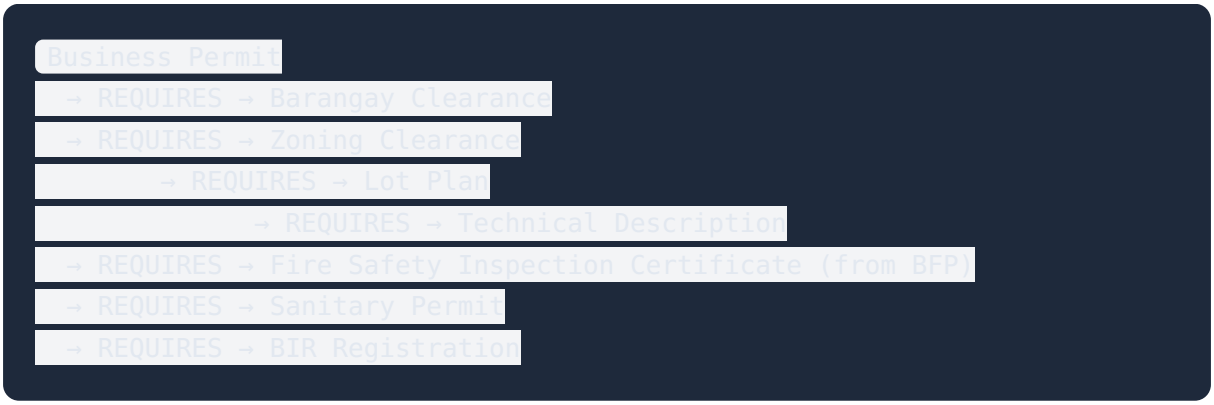
If someone asks "Magkano ang fee noong 2023?", the checkpoint filters out everything that wasn't active in 2023 and returns the 2023 fee — not the current one.

If someone asks about the current fee, it filters out everything with a repeal_date in the past.

The publication lag rule: A law passed on March 15 and published on March 20 is legally effective on April 4 (20 + 15 days = April 4). If someone asks about the fee on March 25, the new fee should NOT be served — it's not legally effective yet. The checkpoint enforces this automatically.

Checkpoint 2: Recursive Requirements Walk — What Do You ACTUALLY Need?

This is the most powerful part of the graph. When someone asks "Ano LAHAT ng requirements ng business permit?", the graph doesn't just return one list. It follows the chain.



The graph walks down every branch, collecting ALL prerequisites. Even prerequisites OF prerequisites. It returns the complete tree, grouped by depth — so the citizen sees:

Safety guards:
- **MAX_DEPTH = 5:** If the chain goes deeper than 5 levels, stop and flag it. Something might be wrong.
- **Cycle detection:** If a requirement points back to something already in the chain (like A requires B which requires A), stop immediately. This prevents infinite loops.

Checkpoint 3: Precedence Resolution — Which Law Wins?

Sometimes two laws say different things about the same fee. Philippine law has a hierarchy:

Tier	Type	Example
0	Republic Act (national law)	RA 7160
1	Ordinance (local law)	Ordinance 2025-012
2	Executive Order	EO 2025-011
3	Administrative Order	AO 2024-005

Lower tier number ALWAYS wins. A Republic Act overrules an Ordinance. An Ordinance overrules an Executive Order.

What if two ordinances (same tier) conflict?
1. The more SPECIFIC one wins (a rule about "restaurants in commercial zones" beats a general rule about "all businesses")
2. If same specificity, the MORE RECENT one wins
3. If same date, flag it as an unresolved conflict and tell the user both exist

Checkpoint 4: Zettel Retrieval — The Fine Print

Once the graph has found the right regulation (like "Building Permit"), it retrieves the granular details stored as "Zettels" — independent mini-documents attached to the regulation:

- **Fee Zettel:** The fee table — how much, for what category
- **Requirements Zettel:** The checklist of documents needed
- **Procedure Zettel:** Step-by-step instructions
- **Penalty Zettel:** Fine amounts for violations
- **Form Zettel:** Links to downloadable application forms

Each Zettel has its own vector embedding, so the system can retrieve exactly the right fee row or requirement without pulling the entire regulation.

Checkpoint 5: Draft Safety Check — Is This Even a Real Law?

Sometimes proposed ordinances appear in session agendas. They look like laws but haven't been passed yet.

This checkpoint asks: "Is this legislation actually enacted?"

- If `enactment_status = "approved"` → Serve it
- If anything else ("draft", "tabled", "vetoed") AND the user is a citizen → BLOCK it. Never tell a citizen to follow a draft.
- If the user is an employee tracking legislation → Allow it, but tag it as `[DRAFT — NOT ENACTED]`

Chapter 4: The Vector Path — How the Search Engine Works

When a question is routed to the vector index, it goes through a simpler but still careful process.

Step 1: Source Quality Filter — Is This Source Trustworthy?

Philippine LGU documents vary wildly in quality:
- Clean 2025 PDFs: Very reliable

- Scanned 2010 documents: Can be misread by OCR
- Photocopied 1998 ordinances: High risk of errors
- Handwritten annotations: Unreliable

Every document chunk has an `extraction_confidence` score (0.0 to 1.0):

Score	Handling
≥ 0.7	Serve normally. High confidence.
0.3–0.7	Serve with a disclaimer: "This source is low quality. Verify with the issuing office."
< 0.3	REFUSE to serve. Don't risk a wrong answer.

The legacy default: Documents ingested before we added confidence scoring default to 0.5 (medium-low confidence) — NOT 1.0 (perfect). We assume old data is imperfect until proven otherwise.

Step 2: Unmapped Document Fallback — Stuff Not in the Graph

The graph can't know everything. There are document types we haven't mapped yet:
- SK (youth council) resolutions
- FDP (Full Disclosure Policy) forms
- ELA (Executive-Legislative Agenda) documents
- Tourism pages, history articles, news posts

When the graph returns empty (meaning "I don't have this mapped"), the vector index covers the gap. It searches the entire document corpus and finds relevant chunks — even for document types that don't fit our 14 node types.

Step 3: Parent-Child Chunking — Tables Done Right

LGU documents often contain complex tables — fee schedules, tax brackets, zoning classifications. A standard text splitter would break a fee table like:

```
| Residential | ≤100 sqm | ₱5,000 |
| Commercial  | ≤100 sqm | ₱7,500 |
```

...into disconnected fragments. The vector index uses **parent-child chunking**: each table row becomes its own searchable chunk, but carries the header context:

```
"Building Permit Fee Schedule → Residential → ≤100 sqm: ₱5,000"
"Building Permit Fee Schedule → Commercial → ≤100 sqm: ₱7,500"
```

Now when someone asks "Magkano ang building permit fee for commercial property?", the vector index directly finds the exact row, not just a page that happens to contain a fee table somewhere.

Chapter 5: The Merger — When Both Brains Report Back

Sometimes both the graph and the vector index return results. The merger decides what to use.

Rule 1: Graph Wins on Overlap

If both brains return information about the same thing (e.g., the building permit fee), the graph's version is authoritative. The vector chunk is partially merged:

- **Graph structural fields** (fee amount, legal basis, effective date): KEPT
- **Vector structural fields** (fee amount, legal basis from an old PDF): DISCARDED
- **Vector narrative content** (office hours, processing tips, "proceed to Window 3"): PRESERVED

Think of it like this: the graph provides the skeleton (the authoritative facts), and the vector provides the flesh (the helpful context around those facts).

How "Overlap" Is Detected

The merger uses three checks, from most to least confident:

1. **Account code match:** Both reference the same government accounting code (e.g., "40201-010-02") → DEFINITE overlap

2. **Synonym match:** Both reference the same entity via our synonym dictionary ("Building Permit" = "Permit to Construct") → HIGH confidence overlap
3. **Semantic similarity:** The text embeddings are more than 90% similar → PROBABLE overlap (least confident, used as last resort)

Rule 2: Conflicts Are Flagged, Not Silently Resolved

If the vector chunk says ₱3,500 and the graph says ₱5,000, the merger does NOT pick one. It flags the discrepancy:

```
[CONFLICT FLAGGED]
Graph shows: ₱5,000 under Ordinance 2025-012
One document states: ₱3,500 — may be from Ordinance 2018-045 (superseded)
```

The LLM (Riva's language model) receives both and can explain the discrepancy to the user: "The current fee is ₱5,000. Some older documents still reference ₱3,500, but this was updated in 2025."

How conflicts are detected:
- **Numeric conflicts:** Any number in the vector chunk that differs from the graph value by more than 10% is a conflict
- **Semantic conflicts:** Keywords like "not required," "no longer," "exempt," "waived" in the vector chunk that contradict the graph's requirements

The Temporal Reconciliation Pre-Check

Before the merger even runs, a **Temporal Reconciliation Engine** scans vector chunks and checks: does the graph have a newer version of this same information?

If the graph node has an effective_date of 2026 and the vector chunk is from a document dated 2024, the vector chunk is PRUNED — removed entirely. No conflict flag. No partial merge. Just gone.

This prevents the most dangerous failure mode: a citizen getting old information from a vector search when the graph has the current version. Think of it like the librarian checking publication dates before letting the search engine results through.

Chapter 6: Context Assembly — The Final Package

All the verified information — graph data, vector supplements, conflict flags — is assembled into a structured prompt for the LLM.

The prompt looks like this:

```
[GRAPH_VERIFIED]
Fee: ₱5,000.00
Authorized by: Ordinance 2025-012, Section 4
Effective: January 1, 2026

[GRAPH_VERIFIED]
Requirements:
  → Barangay Clearance
  → Zoning Clearance (C-2 zone)
  → Fire Safety Inspection Certificate (BFP)
  → Sanitary Permit
  → Electrical Permit

[VECTOR_SUPPLEMENTAL — confidence: 0.85]
"Processing typically takes 7-14 working days.
The City Engineer's Office is open Monday-Friday, 8AM-5PM."

[HAS_PROCEDURAL_NOTE]
"Proceed to Window 3 to verify your tax assessment before payment."

[LOW CONFIDENCE — 0.45, scanned PDF]
Note: One source document was a photocopied 2010 ordinance.
Fee may be ₱4,800–₱5,200. Verify with the Office of the Building Official.

[SYSTEM: Graph database temporarily unavailable]
Answers are from document search only.
```

Why the tags matter: The LLM knows exactly what it's looking at. It knows what's verified, what's supplemental, what's low-confidence, and what's in conflict. It never gets raw blended context where it can't tell which source said what.

Token Budget

LLMs have a limit on how much text they can process at once. The context assembly enforces a 3,500-token budget with priorities:

1. **Graph-verified data** (highest priority — essential)
2. **Conflict flags** (user needs to know about disagreements)
3. **Low-confidence disclaimers** (user needs uncertainty warnings)
4. **Vector supplemental context** (nice to have, first to be truncated if budget exceeded)

If the budget is exceeded, the system adds: `[CONTEXT_TRUNCATED — some information was omitted. Ask for more details.]`

Horizontal Token Guard

Some queries could return dozens of prerequisites. "Ano LAHAT ng requirements?" for a complex permit might have 15-20 items. If each item includes legal basis text, the context explodes. The system caps at **12 branches** — if there are more, it truncates and tells the user to ask for specifics.

Post-Generation Citation Check

After the LLM generates a response, the system checks: did it cite any ordinance numbers? If so, are those ordinance numbers actually in the [GRAPH_VERIFIED] section?

If the LLM cited Ordinance 2025-012 as the source for processing time (which came from a vector chunk), the system appends a correction: "Note: Ordinance 2025-012 is the source for the fee, not the processing time. Verify processing time independently."

Chapter 7: The Complete Journey — Four Examples

Example 1: "Magkano building permit fee?"

```
1. SYNONYM: "building permit" → matched entity "Building Permit"

2. CLASSIFY: structural=0.95 (magkano + entity match), narrative=0.00
   → Route: GRAPH PRIMARY

3. GRAPH PATH:
   Checkpoint 0: KuzuDB is up ✓
   Checkpoint 1: Building Permit effective 2026-01-01, repeal_date=null → valid ✓
   Checkpoint 2: No requirements question, skip traversal
   Checkpoint 4: Zettel(fee) → ₱5,000 ✓
   Checkpoint 5: authorized by Ordinance 2025-012 (approved) ✓

4. VECTOR: Skipped (graph has full coverage)

5. CONTEXT: [GRAPH_VERIFIED] ₱5,000. Ordinance 2025-012.

6. RIVA: "Ang building permit fee ay ₱5,000 base sa Ordinance 2025-012, Section 4."
```

Example 2: "Bakit kailangan ng barangay clearance?"

```
1. SYNONYM: "barangay clearance" → matched entity "Barangay Clearance"

2. CLASSIFY: structural=0.15, narrative=0.85 (bakit)
   → Route: VECTOR PRIMARY

3. VECTOR PATH:
   Step 1: Source quality → chunks at confidence 0.85 (clean PDFs) ✓
   Step 2: Top-5 chunks → Citizen's Charter text, LGC explanation
   → Returns: "RA 7160 Section 152(c) requires barangay clearance as
               verification that the business is compliant with local regulations..."

4. GRAPH: Supplemental — REGULATION "Barangay Clearance" → AUTHORIZED_BY →
          LEGISLATION (RA 7160, Sec 152c)

5. CONTEXT: [VECTOR_SUPP] Rationale text.
            [GRAPH_VERIFIED] Legal basis: RA 7160, Section 152(c).
```

```
6. RIVA: "Ang Barangay Clearance ay kailangan para ma-verify ng barangay
       na compliant ang negosyo sa local regulations. Ito ay base sa
       RA 7160 Section 152(c)..."
```

Example 3: "hm po bgy clearance d2 s naga?" (Text-speak)

```
1. SYNONYM:
   "hm" → "how much"
   "po" → filtered (politeness marker, removed)
   "bgy" → "barangay"
   "d2" → "here"
   "s" → "sa"
   Result: "how much barangay clearance here sa naga?"
   Matched: "Barangay Clearance"
|
2. CLASSIFY: structural=0.70 (hm → "how much" + entity match), narrative=0.00
   → Route: GRAPH PRIMARY
|
[Same as Example 1 from here — the text-speak was transparently resolved]
```

Example 4: "Bawal ba plastic cellophanes sa palengke?" (Temporal conflict)

```
1. SYNONYM: No direct entity match → fuzzy fallback
|
2. CLASSIFY: structural=0.45, narrative=0.35
   → Route: BOTH
|
3. GRAPH PATH:
   Returns: 2026 Administrative Memo — allows plastic for fresh meat packaging
   effective: 2026-03-01, status: approved
|
4. VECTOR PATH:
   Returns: 2024 Ordinance chunk — total ban, no exceptions
   source date: 2024-06-15
|
5. TEMPORAL RECONCILIATION:
   Vector chunk source_date (2024-06-15) < Graph effective_date (2026-03-01)?
   YES → Vector chunk is OBSOLETE. PRUNED.
|
6. CONTEXT: [GRAPH VERIFIED] 2026 memo allows exceptions for fresh meat.
|
```

```
7. RIVA: "Base sa 2026 administrative memo, pinapayagan ang plastic
         cellophanes para sa sariwang karne sa palengke."
```

Chapter 8: The Safety Net — What Happens When Things Go Wrong

No system is perfect. Janus RAG has multiple layers of protection:

KuzuDB (Graph Database) Goes Down

```
TRY: Connect to KuzuDB → FAIL (connection timeout)
FALLBACK: Route ALL queries to vector-primary
CONTEXT TAG: [SYSTEM: Graph database temporarily unavailable.
              Answers are from document search only.
              Verify all fees and requirements with the issuing office.]
```

The system continues working — just with a disclaimer that answers aren't as precise.

ChromaDB (Vector Database) Goes Down

```
TRY: Connect to ChromaDB → FAIL
FALLBACK: Graph-only operation. Vector supplements unavailable.
GAP: Narrative queries ("bakit?", "history") return basic graph data only.
```

Extraction Confidence Is Low

A scanned 1998 ordinance with confidence 0.25:
- Graph: Node is stored but flagged as low confidence
- Vector: Chunk is REFUSED (below 0.3 threshold)
- Context: [LOW_CONFIDENCE — 0.25] with instruction to verify

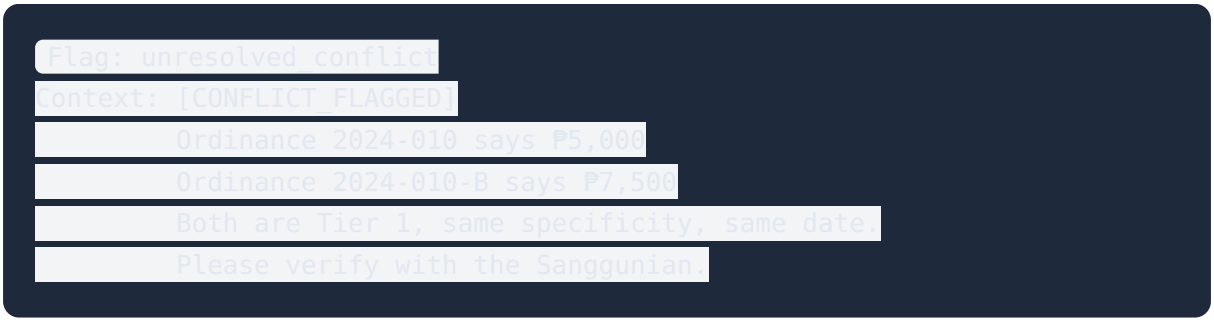
The system never silently serves unreliable data as fact.

Requirements Chain Has a Circular Loop

If Barangay Clearance → REQUIRES → Resident ID → REQUIRES → Barangay Clearance:

- Cycle detection activates at the second visit to Barangay Clearance
- Traversal halts immediately
- Flag: `circular_dependency_detected`
- The citizen gets whatever was collected before the loop (which is usually complete)

Two Ordinances Conflict, Same Tier, Same Date



The system admits it can't resolve the conflict rather than guessing.

Chapter 9: Why "Janus"?

Janus is the Roman god of doorways, gates, and transitions. He has **two faces** — one looking forward, one looking backward. Both serve as gateways.

Our Janus has:

- **The graph face** — looking at structured, precise, legally-grounded information
- **The vector face** — looking at broad, contextual, narrative information

Both faces serve the same user. Both are gateways to the same answer. Neither is a backup for the other — they are complementary, each covering the other's blind spots.

When someone asks "Magkano ang building permit fee?", the graph face answers with precision. When someone asks "Bakit kailangan ng zoning?", the vector face provides context.

And when both faces see the same question, they don't argue — the graph is authoritative, and the vector enriches.

Chapter 10: Proving It Works — The Ablation Experiment

> "Does the hybrid graph+vector architecture actually outperform simpler approaches? Or are we paying dual-database complexity for marginal gains?"

On June 7, 2026, we ran a controlled experiment to answer this definitively. Here's what we did and what we found.

The Setup

Three pipelines, same 4 synthetic LGU documents, same 36 questions:

Pipeline	What It Uses	Description
Vector-Only	ChromaDB + BGE-small	Search engine only — no graph
Graph-Only	KuzuDB graph	Librarian only — no search engine
Janus V4	Graph + Vector	Both brains — the hybrid

The 4 documents: Revenue Code (BP fees, Barangay fees, Sanitary Permit), Zoning Ordinance (R-1/C-2/I-1 zones), Tricycle Franchise (MTOP requirements + penalties), Amendment Chain (Ordinances 2015→2018→2024).

The 36 questions covered 8 categories: Fee Lookup (6), Multi-Hop Reasoning (4), Conceptual "Why" (4), Temporal/Historical (4), Colloquial Filipino (6), Out-of-Corpus (4), Aggregation (4), and Easy Baseline (4).

Every question had a ground truth answer written before running the pipelines. Two independent evaluators — Claude and Gemini — scored blindly (they didn't know which pipeline produced which answer).

The Results

Metric	Vector-Only	Graph-Only	Janus V4
Overall score (Claude)	81.6%	71.3%	**88.4%**
Answer rate	56%	50%	**66%**
Numeric accuracy	65%	50%	**75%**
Fee lookup (6 questions)	93.3%	93.3%	**100%**
Multi-hop reasoning	70%	77%	**95%**
Aggregation	81%	55%	**98%**
Out-of-corpus honesty	100%	100%	100%

What the Numbers Mean

Janus beats either single-pipeline approach. +6.7pp over vector-only, +17.1pp over graph-only. On fees — the most important category for citizens — Janus scored 100% (6/6 answered, 6/6 numeric values correct).

Neither alternative is safe alone. Graph-only cannot answer "why" questions — it scored 44% on conceptual queries because it stores facts but not explanations. Vector-only confidently lies on binary yes/no questions — it told citizens "no Barangay Clearance needed for Building Permit" when the document clearly states ₱50 is required.

Janus covers both blind spots. When the graph can't retrieve narrative text (like "why is zoning clearance required?"), the vector path supplies the explanation. When the vector hallucinates a gap (like "walang nakasaad" for Barangay Clearance), the graph's structured nodes catch it.

The hybrid is graph-first, not vector-first. Janus averages 10.8 graph chunks vs only 2.8 vector chunks per query. Structured data does the heavy lifting. Vector is reinforcement, not the foundation.

Three Independent Evaluators Agreed

Evaluator	Version	Verdict
Claude	v1 (initial)	□ Architecture not justified — but identified 3 specific bugs
Gemini	v2 (bugs fixed)	□ Both hypotheses VERIFIED
Claude	v3 (P0/P1 fixes)	□ Janus wins overall. Architecture validated.

The trajectory matters. Claude's initial "not justified" was correct for the buggy implementation. After fixing the bugs (Office retrieval, stale Zettel handling, temporal interval containment), both evaluators converged on the same conclusion: **the hybrid architecture is necessary and performant.**

What This Means for Production

The remaining ceiling is data, not architecture. Janus scores ~66% answer rate and ~75% numeric accuracy on 4 documents. Claude's analysis: "The 11 in-corpus questions Janus still fails to answer point to corpus gaps more than pipeline architecture issues."

With 1,700+ real Naga City documents — messy amendment chains, cross-referenced ordinances, conflicting dates — the graph's structural advantage compounds. Vector-only degrades on messy data. Janus doesn't.

The dual-database complexity is justified. Not because Janus scores slightly higher, but because it prevents the two worst things an AI government assistant can do:

1. **Tell a citizen "no" when the requirement exists** (Vector-Only's M3 failure)
2. **Say "I don't know" when the explanation is available** (Graph-Only's C1 failure)

Neither failure mode survives the hybrid.

Chapter 11: Glossary

B3 Router: The original intent router from the Thermopylae project. Named because it classified queries into three types (fee, requirement, procedure). Now integrated into the two-dimensional Intent Classifier.

BGE-small: A compact AI model that converts text into number lists (embeddings). Small enough to run on a laptop. Used by the vector path.

ChromaDB: A database for storing and searching vector embeddings. Think of it as a specialized search engine that finds text by meaning, not keywords.

Embedding: A list of numbers (typically 384 or 768) that represents what a piece of text is about. Similar texts have similar number patterns. Used for semantic search.

Extraction Confidence: A score from 0.0 to 1.0 indicating how reliable a document extraction was. Clean PDF = 1.0. Scanned photocopy = 0.4. Handwritten = 0.2.

GraphRAG: Retrieval-Augmented Generation using a knowledge graph. The system retrieves information from a structured graph database (rather than just searching text) and feeds it to an LLM.

Intent Classifier: The component that decides whether a query goes to the graph or vector path. Uses keyword scoring, not an AI model — 0ms latency.

KuzuDB: An open-source graph database. Stores nodes and edges. Used by the graph path. Like Neo4j but embeddable (no separate server needed).

LLM: Large Language Model — the AI that generates the final response. Riva uses a model like Qwen3 or DeepSeek to write answers in Filipino/English.

MAX_DEPTH: The maximum allowed depth for recursive requirement traversal. Set to 5. Deeper chains are suspicious and halted.

Overlap: When the graph and vector index both return information about the same entity. Detected by account code match, synonym match, or semantic similarity.

PRECEDENCE_TIER: A legal hierarchy for resolving conflicts between laws. Republic Act (0) > Ordinance (1) > Executive Order (2) > Administrative Order (3).

Publication Lag: Philippine laws take effect 15 days after official publication (RA 7160, Section 59). The temporal filter enforces this automatically.

REQUIRES Chain: The chain of prerequisites for a permit. "Building Permit REQUIRES Zoning Clearance REQUIRES Lot Plan" = a 3-hop chain. The graph traverses this recursively.

SUPERSEDES: An edge indicating that one legislation fully replaces another. When detected, all child regulations of the old legislation get their repeal_date set automatically.

Synonym Alignment Interface: Pre-processor that maps colloquial/regional terms to canonical graph entity names. Handles "bgy clearance" → "Barangay Clearance", text-speak, and Bikol-influenced terms.

Temporal Reconciliation: Pre-merger check that prunes vector chunks from older documents when the graph has a newer superseding version. Prevents stale data from contaminating answers.

Zettel: An independent mini-document node stored in the graph with its own vector embedding. Contains granular content like fee tables, requirement checklists, or penalty schedules. Connected to parent nodes via HAS_ZETTEL edges.